

Evaluation and Human-in-the-Loop Annotation Using GenAI Approaches

A Reproducible Framework for Classification, Document-Scale Extraction, and Reviewed Gold Data

Amine Smaoui · Ismail Ege Oguz · Robert Pham

School of Computation, Information and Technology, Technical University of Munich

NLP Lab, Summer Semester 2026. Supervised by Dr. Georg Dietrich and Dr. Maximilian Wich, Carl Zeiss AG.

Abstract. Large language models (LLMs) make information extraction and ticket classification available without task-specific training, but useful deployment still requires evidence on the target data and an efficient way to create that evidence. We present a reproducible evaluation framework and a configurable human-in-the-loop annotation application. The framework preserves raw model responses, validates spans and labels, and reports strict and reviewer-oriented metrics. The application suggests a schema-derived prompt, pre-annotates text, PDF, or CSV input, supports optional department routing through either an editable LLM prompt or a local embedding vote over reviewed tickets, records every reviewer correction, and exports traceable gold JSON. On a fresh held-out sample of 20 SciREX papers, four-type scientific mention extraction reached 57.38% strict F_1 (document-bootstrap 95% interval 53.56–60.73), 67.36% boundary-tolerant F_1 , and 70.30% same-label overlap F_1 . A completed 100-paper full-window operational run processed 8,373 sentences and 9,901 gold mentions with no failed document, achieving 54.09% strict and 66.22% overlap F_1 . Earlier baselines show 79.32% strict F_1 for single-type Few-NERD extraction and 72.6% type accuracy but only 42.2% queue accuracy on synthetic support tickets. The results show that exact span quality, label semantics, and target-domain evaluation matter more than output validity alone, and that LLM predictions should remain reviewer suggestions rather than ground truth.

Keywords: information extraction; text classification; large language models; long documents; evaluation framework; human-in-the-loop annotation; reproducibility

1 Introduction

Information extraction (IE) and text classification turn unstructured text into structured, usable data. A service engineer repairs a machine and records the customer request, the observed failure, and the corrective action. Across thousands of reports, these records contain information required for quality analysis in a form that conventional dashboards cannot directly process. IE extracts the relevant facts, whereas classification assigns each document to a category [1].

Until recently, this process required substantial initial investment, workflow (a) in Figure 1: domain experts labeled thousands of examples, machine-learning specialists trained and tuned a model, and label-set changes could require additional annotation and training. Instruction-tuned large language models (LLMs) reduce this initial effort because a task can be specified through label descriptions and examples in a prompt, workflow (b).

The principal limitation is the loss of independent evaluation data. The labels created for supervised learning also served as a test set against which predictions could be measured. Organizations may therefore lack annotations for their target domain and cannot reliably evaluate deployed models. Vendor benchmarks may differ substantially from service reports, while informal spot checks cannot establish deployment quality. LLM annotation quality varies across tasks and datasets and must therefore be measured against human labels rather than assumed [2, 3].

Workflow (c) is implemented in this study. An LLM pre-annotates, a domain expert validates, and the result is a small gold set to measure against. We built both halves: an evaluation framework for classification and span extraction, and a configurable annotation product that turns model output into reviewed data. The initial Few-NERD and support-ticket experiments test method and output choices on short inputs. The later SciREX study addresses the harder operational case: coherent scientific text ranging from one sentence to more than 200, processed completely with durable checkpoints and evaluated against released span annotations [4]. This progression matters because sentence-local predictions that work on isolated inputs must still be shown to cover every sentence in a full document reliably and economically.

Contributions. Our contributions are threefold. First, we provide a task-separated, configuration-driven evaluation framework that stores raw responses and scores them independently of inference. Second, we construct a reproducible 1,000-window SciREX benchmark from 438 source papers while preserving source order, document identity, split membership, and exact character offsets; the completed study stages include a development prompt comparison, a fresh 20-paper held-out confirmation, and a 100-paper full-window operational run. Third, we deliver a Dockerized annotation application with schema-derived but editable prompts, arbitrary labels, text/PDF/CSV input, optional reviewer-approved LLM or local-embedding ticket routing, confidence-

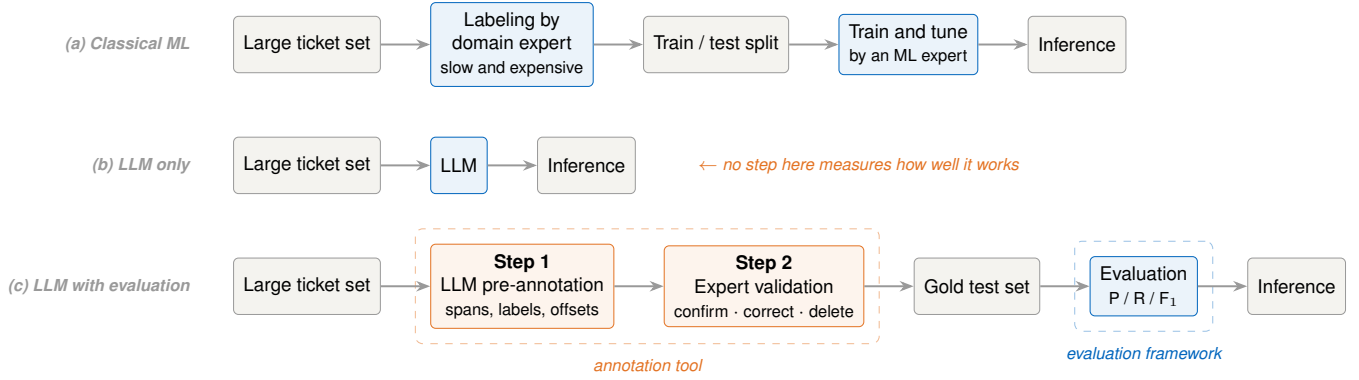


Figure 1: Three annotation workflows. (a) The supervised pipeline requires expert labeling before model development. (b) Direct LLM inference omits an independently labeled evaluation set. (c) The implemented workflow combines LLM pre-annotation with expert validation and gold-set evaluation. Dashed boxes denote the two artifacts developed in this project.

ranked review, and auditable JSON export. Together, these artifacts define a reproducible process from unlabeled organizational text to an evaluation set with inspectable provenance.

2 State of the Art and Thematic Analysis

We read work published between 2018 and 2025, mostly peer-reviewed and from the main NLP and machine-learning venues (ACL, EMNLP, NAACL, EACL, COLING, ICLR, NeurIPS), from reviewed journals (*PNAS*, *Computational Linguistics*, *Frontiers of Computer Science*, *JMIR Medical Informatics*), or from CHI and WWW for the human-in-the-loop side. Seven sources are arXiv preprints, kept only where the work is widely cited and from an identifiable group, and marked as preprints rather than treated as reviewed. Sources had to speak to one of four questions: how well prompted models extract and classify, how their output should be shaped, how a reviewer should be pointed at the doubtful cases, and how any of it can be checked without gold data. Eight themes follow.

From supervised pipelines to prompted generalists. Xu et al. [1] survey generative IE. Generative formulations can accommodate dynamically specified labels through natural-language descriptions, avoiding a fixed classifier output layer. Specialized instruction-tuned systems such as InstructUIE [5] and targeted-distillation systems such as UniversalNER [6] demonstrate that task-specialized smaller models can outperform general-purpose prompted LLMs on IE benchmarks. GLiNER [7] makes a related point from the encoder side: a bidirectional encoder matches LLM prompting at a fraction of the inference cost. **In our project**, fine-tuning was out of scope: we had access to one hosted model and no permission to train on company data. So we compare only methods that need no training step.

How far prompting alone gets on extraction. Han et al. [8] ask whether IE is solved by ChatGPT and answer no: performance holds on tasks rewarding reasoning and drops on sequence labeling, where boundaries count. Xie et al. [9] show that decomposing NER into simpler label-specific questions can substantially improve zero-shot performance. Sivarajkumar et al. [10] compare six prompt types across five clinical tasks and find no overall winner, so prompt design must be tested per task. GPT-NER [11] adds a self-verification step asking whether an extracted span really belongs to the target type. **In our project**, two of these

claims are testable, so we test both (H1, H3).

Document-level scientific extraction. SciREX was introduced specifically because sentence- and paragraph-level IE misses information distributed across scientific documents [4]. It annotates mentions of Method, Task, Metric and Dataset as well as coreference, salience and document-level relations. Our study uses its mention spans as externally supplied reference annotations; it does not attempt the original paper’s coreference or n -ary relation tasks. This distinction prevents a misleading direct comparison between our pre-annotation score and the SciREX end-to-end relation baseline.

Classification without training data, and label-set size. Yin et al. [12] turn classification into natural language inference with each label as a hypothesis, which works for any label set at inference time but reacts strongly to wording. The embedding alternative encodes text and label descriptions and selects the nearest representation [13]; it is computationally inexpensive, local, and requires no LLM call, and SetFit [14] shows how much it gains from eight labeled examples per class. MTEB [15] is the usual way to choose an encoder. The harder problem is scale: Bhambharia et al. [16] show that flat zero-shot classification degrades on layered label sets and that splitting the decision into entailment steps recovers much of the loss, TELEClass [17] adds class-indicative terms to the label set, and HierPrompt [18] builds hierarchy-aware prototypes. All three assume that putting hundreds of labels in one prompt does not work. **In our project**, the label sets have 4, 10 and several hundred values, so we can watch this drop happen inside one dataset (H2).

Output format is not a free choice. Tam et al. [19] show format restrictions measurably hurt reasoning, and tighter constraints hurt more, though structured formats *help* classification, where the task is choosing rather than reasoning. Span extraction additionally needs offsets consistent with the emitted text, which no JSON schema checks. **In our project**, each extraction method exists twice, structured and free-form, with accuracy and structural validity measured separately.

Whether an agentic step helps. Huang et al. [20] study intrinsic self-correction and find performance usually does not improve and can get worse, because the model has no independent signal about which parts were wrong. Work on

LLMs as judges documents position, verbosity and self-preference biases pointing the same way [21]. **In our project**, we treated “agentic helps” as a hypothesis to test rather than assume, and built two agentic variants for it (H3).

LLMs as annotators, and why validation is not optional.

Gilardi et al. [22] report ChatGPT beating crowd workers on several annotation tasks at a fraction of the cost. The careful work came soon after. Ziems et al. [23] test 13 models on 25 benchmarks and find fair agreement with human coders but no win over fine-tuned models, Reiss [3] shows non-determinism alone can push consistency below accepted reliability thresholds, and Pangakis et al. [2] replicate 27 tasks across 11 datasets and conclude every task needs its own check against human labels. Baumann et al. [24] quantify the downstream risk: across 13 million annotations feeding 1.4 million regressions, ordinary choices of model and prompt wording sufficed to flip statistical conclusions. See Tan et al. [25] for a survey. **In our project**, this theme is the reason the project exists. It also fixes one rule for the tool: every human decision gets stored, because the record of disagreement is the evidence.

Prioritizing uncertain cases for review. A reviewer facing forty plausible suggestions stops reading them properly, so the tool must rank them. Geng et al. [26] survey confidence estimation and split it into white-box methods reading internal signals such as token probabilities and black-box methods treating the model as opaque; Kadavath et al. [27] show large models carry usable self-knowledge in their probabilities. The black-box route is sampling: Wang et al. [28] aggregate several sampled generations and Kuhn et al. [29] refine this further by grouping samples that mean the same thing before counting them. Xiong et al. [30] compare the families and report two findings we relied on: models are overconfident when stating confidence in words, and white-box methods beat black-box ones only narrowly. For annotation specifically, Wang et al. [31] study how to make human verification effective rather than mechanical. **In our project**, the annotation tool supports multiple confidence signals for prioritizing review; a systematic comparison of their calibration, review utility, and additional cost remains future work.

Keeping the human in the loop. INCEpTION [32] already paired automatic suggestions with human curation before LLMs; MEGAnno+ [33] splits the work as we do, with LLM agents proposing and people verifying. For annotation quality the standard vocabulary is Artstein and Poesio [34]. **In our project**, we took that two-step design, adapted it to span-level extraction with character offsets, and logged every accept, relabel and delete.

The integration considered here. Existing work studies individual components of this process, but we did not find a single task-separated workflow combining reproducible LLM inference, independently scored extraction and classification, reviewer correction, provenance, and reusable gold-data export for the target setting considered here. This project implements and evaluates that specific integration.

3 Problem Statement and Research Questions

An organization can obtain plausible extraction and classification output from a general LLM without training a task-

specific model. Plausibility is not an accuracy measure, however, and the organization usually lacks reference labels for exactly the documents it wants to process. It must therefore solve two linked engineering problems: evaluate a proposed method on representative data, and create trustworthy annotations without making experts label every document from scratch. Long documents add operational concerns—call volume, partial failures, offset reconstruction and resumability—that short benchmark sentences conceal.

We address the following questions:

- RQ1** How accurately and reliably does a training-free LLM reproduce reference labels for ticket classification and span extraction, under strict structural and semantic validation?
- RQ2** How well does schema-guided pre-annotation localize Method, Task, Metric and Dataset mentions in coherent SciREX text using sentence-local predictions, and how does performance change from short to very long document windows?
- RQ3** Can one reusable human-in-the-loop product turn model predictions into auditable reviewed data across scientific extraction and internal-ticket use cases?

The study is intentionally not a model leaderboard. A single hosted deployment is fixed so that changes in data, prompt and pipeline remain visible. We use strict span F_1 as the primary extraction measure, because a gold export must reproduce exact offsets, and predeclare boundary-tolerant and overlap metrics as secondary measures of whether a suggestion still directs a reviewer to the useful phrase. For document-scale evaluation, source paper rather than window is the statistical unit because several windows may originate from one paper.

Three expectations guide the analysis. First, syntactically valid JSON will not guarantee correct labels or boundaries. Second, schema definitions and explicit negative guidance will materially affect extraction quality. Third, a complete document-scale run should be technically possible through sentence-level checkpointing, but accuracy need not improve with longer windows because calls remain sentence-local. These are evaluated only where retained artifacts support the comparison; unexecuted configurations are not presented as results.

4 Methodology

Experimental control and provenance. All reported LLM runs use the configured `hosted:gpt-5.4` deployment at temperature 0. Each prediction record stores the example and document IDs, model and method, raw and parsed output, prompt and manifest hashes, token usage, elapsed time and validation state. Evaluation is a separate deterministic step, so alternative matching policies can be applied without another API call. We report only conditions backed by retained prediction and score artifacts.

Short-input baselines. Ticket classification predicts one `type`, one `queue` and a set of `tags`. A zero-shot call selects from allowed values and quotes evidence; an agentic two-step method first extracts evidence and then maps it to labels. Both completed the same 1,000-ticket slice. Few-

shot classification is implemented but only smoke-tested on one row and is excluded from comparison. Embedding-based retrieval of reviewed tickets is also implemented as an optional dynamic few-shot method, but no leakage-safe reference/development/test split was available for the reported synthetic-ticket experiment; consequently, it is not included in the reported comparison. For interactive use, the annotation application additionally implements a pure nearest-neighbor router: it embeds a ticket locally and applies a similarity-weighted vote over reviewed reference tickets, without making a routing LLM call. This route is likewise excluded from the reported accuracy results because it has not yet been tested on an independent target-domain split. For Few-NERD, the two directly comparable 1,000-sentence conditions are few-shot structured JSON and few-shot free-form output, each using three training examples and explicit inclusive token indices. The zero-shot extraction artifacts contain 100 examples and are reported only as a supplementary scale check, not as a full-slice comparison.

SciREX prompt selection and frozen evaluation. The application derives a deterministic prompt from the active label schema, definitions and domain guidance; this suggestion costs no LLM call and remains editable. On a fixed 20-paper development sample, we compared the previous static few-shot prompt with this suggestion using the first ten app-detected sentences per window. The selected prompt presents Method, Task, Metric and Dataset to users; Dataset is mapped to SciREX’s release label Material during evaluation. After prompt selection, a fresh 20-paper official-test sample, excluding test papers already inspected, was processed in full. A separate 100-paper training-split gate, balanced across four length buckets, tested full coverage, throughput and failure recovery. No test result was used to revise the reported prompt.

Document-scale execution. Windows are split into sentences only for API calls; all sentences are processed when `max_sentences=0`. Four workers handle independent documents. Before each call and after each successful sentence, state is written atomically. Repeating an identical command resumes at the next unfinished sentence, while hash checks reject mixed manifests, models or prompts. The runner enforces call and request-rate ceilings, retries transient failures, and records missing usage and invalid responses. Provider prices were unavailable, so monetary cost is not inferred from token counts.

Metrics. For extraction, strict one-to-one matching requires the same case-insensitive label and exact half-open character span. Boundary-tolerant matching permits a start within one token and an end within two; overlap matching accepts any same-label character overlap. Strict is primary; the other two quantify useful localization for human review. We report micro precision, recall and F_1 , macro-document F_1 , per-label and length-bucket results. Uncertainty intervals use 2,000 percentile-bootstrap samples clustered by source `doc_id`. Ticket single-label fields use accuracy and macro- F_1 ; multi-label tags use micro- F_1 . Invalid JSON, invalid labels, evidence validity, failures, token usage and coverage are reported as operational measures.

5 Data

No confidential industrial ticket or service-report data is used. The public datasets test mechanics and establish baselines; they do not substitute for an evaluation on the deployment organization’s own distribution.

Synthetic support tickets. The multilingual customer-support collection [35] contains English and German synthetic emails with provided routing fields. Our processed table has 24,743 rows; experiments use a fixed first slice of 1,000. The label spaces contain 4 ticket types, 10 queues, and 1,129 observed tag strings. The tags deliberately remain noisy, including case variants, misspellings and comma-joined values. These are dataset-provided reference labels, not expert annotations from the intended deployment, so results measure reproduction of this synthetic labeling policy.

Few-NERD. Few-NERD [36] contains 188,238 Wikipedia sentences with 8 coarse and 66 fine types. We use a deterministic 1,000-sentence location-only slice from the intra test split for the directly comparable few-shot runs. Token-level reference spans make it a useful regression test for prompt parsing and offset reconstruction, although isolated Wikipedia sentences do not test document-scale execution.

SciREX. SciREX [4] provides complete tokenized scientific papers with Method, Task, Metric and Dataset/Material mention spans, created through a combination of automatic and human annotation. The release used here contains 438 documents, 120,432 sentences and 156,931 mention spans. Preprocessing detokenizes each document deterministically, converts exclusive token offsets to validated half-open character offsets, and preserves the original train/development/test split.

From these papers we construct 1,000 contiguous same-document windows, 250 in each bucket: short (1–10 sentences), medium (11–50), long (51–200), and very long (at least 201). Sentence order is never randomized. Windows from a paper may overlap by at most 50%, so `doc_id` must be retained for splitting and uncertainty estimation. The full benchmark contains 109,138 gold mentions and is available as linked example, sentence and entity CSV tables. The completed 100-paper operational subset contains 25 unique training papers per bucket, 8,373 sentences and 9,901 mentions. The fresh held-out confirmation contains 20 different official-test papers, 1,633 sentences and 2,359 mentions. The configured 1,000-window run remains prepared but was not completed and is therefore not reported as an empirical result.

6 Architecture and Solution Design

Two artifacts implement workflow (c) of Figure 1: an evaluation framework for data with reference labels and an annotation product for creating reviewed labels where none exist.

6.1 Evaluation framework

The Python repository is separated by task. Configuration and prompt files describe a run; method classes expose a common prediction interface; runners handle API calls, retries and validation; evaluators read stored predictions and write aggregate, per-example, per-label and error arti-

facts. The SciREX adapter adds deterministic detokenization, document-window construction and offset assertions. A production batch runner adds atomic sentence checkpoints, configuration hashes, bounded parallelism, rate and call ceilings, token accounting and idempotent resume. This separation is important: inference is expensive and stateful, whereas scoring should be computationally inexpensive, inspectable, and repeatable.

For ticket classification, an optional dense-retrieval component indexes reviewed reference tickets with a pretrained sentence encoder and supplies the nearest valid examples to the LLM prompt. The index manifest stores source and artifact hashes, encoder identity, dimensions and column mappings. Reference and evaluation data remain separate, and query-time exclusion by ticket ID and normalized text provides an additional leakage safeguard. This component is a retrieval baseline, not LLM fine-tuning, and remains excluded from reported results until it is evaluated on an independent internal split.

6.2 Human-in-the-loop annotation product

The Streamlit application accepts pasted text, TXT, PDF, or an interactively mapped CSV row. A user selects or copies a YAML use-case template, edits labels and the complete schema-derived prompt, chooses an OpenAI-compatible or local Ollama model, and runs pre-annotation. Suggestions are highlighted in context and listed with exact character offsets, provenance, status and confidence (Figure 2). Reviewers can confirm, relabel, delete, adjust or add spans. The export retains immutable model predictions alongside final `gold_entities`, prompt hashes, token usage and the full review history.

Reusable domain configuration. YAML templates currently cover general named entities, scientific IE and internal support tickets. Each defines labels, descriptions, domain guidance and optional routing destinations. The prompt suggestion is deterministic, so it is reproducible and free, but the user can adapt it before inference. Custom labels receive a generic starting definition instead of disabling the tool.

Optional classification before extraction. For tickets, the user selects either an LLM classifier or a local embedding classifier. The LLM route uses a fully editable prompt, selects one department from the allowed list and quotes ticket evidence. The local route maps a reviewed CSV into ticket, department and ID columns, builds a pretrained sentence-embedding index, retrieves the nearest reviewed tickets and applies a similarity-weighted vote. It makes no routing LLM call, although extraction can still use an LLM. The export records the classifier, prediction, confidence, review decision and, for local routing, the encoder, source hash and neighbor provenance. The reviewer confirms or corrects routing independently of entity review. The evaluation page can summarize reviewer acceptance by classifier, but this workflow measure is not independent classification accuracy; accuracy requires labels assigned without seeing the prediction.

Long text and provenance. The tool processes sentences sequentially and converts their local spans to document-level offsets. The user may choose a sentence limit for a quick test or process the complete document. Sentence-

Table 1: Verified ticket-classification results ($n = 1000$). Accuracy and macro- F_1 (mF_1) are shown for single-label fields; tags use micro- F_1 .

Method	Type		Queue		Tags
	Acc.	mF_1	Acc.	mF_1	F_1
Zero-shot	.726	.606	.422	.366	.291
Agent two-step	.719	.632	.420	.363	.268

local calls reduce context size and make failure recovery precise, but they cannot resolve entities requiring cross-sentence context. The batch runner addresses scale through checkpoints rather than silently truncating documents.

Distribution and security boundary. A pinned Python 3.11 image, Dockerfile and Compose file make the product independent of a particular IDE; a Dev Container remains optional. Credentials enter at runtime through ignored environment or Streamlit-secret files and are excluded from the Docker build context. Reviewed exports are mounted to the host. This package is reproducible for evaluation, but it is not yet a multi-user production service: authentication, role-based access, central persistence, queues and tenant isolation remain outside the current implementation.

7 Results

7.1 Verified short-input baselines

Table 1 contains only rows whose saved score files match the stated sample size. On synthetic ticket classification, a second evidence-extraction call did not improve accuracy: zero-shot and agent two-step differ by 0.007 on type and 0.002 on queue, while zero-shot has the higher tag F_1 . Both produced valid JSON and allowed labels for every example; their quoted evidence occurred in the input in 95.28% and 94.84% of cases, respectively.

For location-only Few-NERD extraction, few-shot structured JSON reached precision 0.7875, recall 0.7990 and strict $F_1 = 0.7932$ on 1,000 sentences. Few-shot free-form output reached 0.7892, 0.7933 and 0.7913. The 0.0019 difference does not establish a format winner. On a separate 100-sentence slice, zero-shot structured and free-form reached 0.8080 and 0.8030 strict F_1 . An earlier 66-label prompt lacking an indexed token table reached only 0.2993 strict versus 0.6070 lenient F_1 ($n = 1000$), although JSON validity was 1.0. Since both label scope and offset instruction changed, this diagnoses a configuration failure but does not isolate a causal prompt effect.

7.2 SciREX prompt selection and held-out confirmation

On the fixed 20-paper development sample, replacing the static few-shot prompt with the deterministic schema suggestion raised strict F_1 from 0.4875 to 0.6319 (Table 2). This comparison was used for prompt selection and is not held-out evidence. The selected prompt then processed a fresh, disjoint set of 20 official-test papers in full. Strict F_1 was 0.5738, with a document-bootstrap 95% interval of 0.5356–0.6073; boundary-tolerant and overlap F_1 were 0.6736 and 0.7030. All 1,224 calls completed, with no failed document or invalid structured response.

On the fresh held-out sample, strict label F_1 was 0.5983 for Method, 0.5739 for Dataset, 0.5309 for Task and 0.4874

Text	Label	Start	End	Source	Status	Confirmed	Relabeled
Munich	location	40	46	gpt-5.4	pending	☐	☐
DHL Hub	building	67	74	gpt-5.4	pending	☐	☐
John Miller	person	91	102	gpt-5.4	pending	☐	☐
Berlin	location	155	161	gpt-5.4	pending	☐	☐
Amazon Logistics	organization	201	217	gpt-5.4	pending	☐	☐
Frankfurt warehouse	building	243	262	gpt-5.4	pending	☐	☐

Figure 2: Human review in the annotation product. Model spans are highlighted in the source text and exposed as editable records; corrections and deletions remain part of the audit trail rather than being overwritten.

Table 2: SciREX prompt development and fresh full-window confirmation. Flexible metrics were introduced with the schema-derived prompt.

Condition	<i>n</i>	P	R	Strict	Ov. F_1
Static prompt, dev prefixes	20	.424	.573	.488	—
Schema prompt, dev prefixes	20	.657	.609	.632	.742
Schema prompt, held-out full	20	.596	.553	.574	.703

Table 3: Strict SciREX results by full-window length in the 100-paper operational run. Each row contains 25 unique training papers.

Bucket	P	R	F_1	Overlap F_1
Short (1–10)	.576	.632	.603	.665
Medium (11–50)	.537	.567	.552	.665
Long (51–200)	.553	.515	.533	.668
Very long (≥ 201)	.539	.544	.541	.659
All	.544	.538	.541	.662

for Metric. The difference between strict and overlap scores shows that a substantial share of errors are boundary disagreements rather than completely irrelevant suggestions. It does not make overlap a replacement for exact scoring: exported training data still requires the reviewer to correct those boundaries.

7.3 100-paper full-window operational run

The operational gate completed all 100 training papers, balanced 25 per length bucket, and processed all 8,373 sentences and 9,901 reference mentions. Strict micro F_1 was 0.5409 (95% paper-clustered interval 0.5190–0.5622), boundary-tolerant F_1 0.6301, and overlap F_1 0.6622. Macro-document F_1 was 0.5065.

Strict performance is lower outside the short bucket but does not decline monotonically with window length; overlap F_1 stays within 0.009 across all four buckets. This is evidence that the runner covers long windows reliably, not that it uses document-level context: predictions are still made one sentence at a time.

The run made 5,868 logical calls in 12,561.81 seconds, consuming 2,645,585 input and 260,276 output tokens. No document failed, coverage was 100%, and one structured response was invalid. The recorded cost is not reported as

zero: provider prices were not configured, so a monetary estimate would be unknown. Per-label strict F_1 was 0.5919 for Method, 0.4944 for Task, 0.4890 for Metric and 0.3220 for Dataset, making Dataset overprediction the largest remaining label-specific problem. The prepared 1,000-window configuration would require exactly 62,145 logical calls, but it has not been completed and is not presented as a result.

8 Discussion

RQ1: reliable output is not the same as valid output. The short-input baselines demonstrate two different ceilings. Single-type Few-NERD extraction reaches about 0.79 strict F_1 , while synthetic-ticket routing reaches only 0.422 accuracy despite perfect JSON and allowed-label validity. The agentic classifier adds a call but does not improve accuracy. The proper conclusion is narrow: on this model, prompt and dataset, extra intrinsic reasoning is not evidence of better predictions, consistent with findings on self-correction without external feedback [20]. The unclear 1,129-tag vocabulary and synthetic origin of the ticket corpus also mean this is a stress test, not an estimate for an internal ticket system.

RQ2: useful pre-annotation, with a substantial review burden. The fresh SciREX confirmation gives the cleanest quality estimate: 0.5738 strict and 0.7030 overlap F_1 . Thus the model often points to a relevant phrase without reproducing the annotation boundary exactly. That distinction matches the product goal. A reviewer can benefit from useful localization, but the output is not yet suitable as unattended training data. Metric and Dataset remain the hardest labels, and the 100-paper run shows Dataset precision of only 0.215 despite recall of 0.642. Better negative definitions and reviewed in-context examples should target this error before changing models.

The length analysis is encouraging operationally. All full windows, including at least 201 sentences in the very-long bucket, completed and overlap F_1 was stable. It does not show that more document context helps because each sentence is predicted independently. Rather, it shows that length no longer causes silent truncation or unrecoverable execution. Cross-sentence coreference and relation extraction remain separate problems, and our mention score is not directly comparable with SciREX’s original document-level relation pipeline [4].

RQ3: a reusable path to gold data is implemented, not yet validated as a productivity gain. The product now covers configuration, prompt inspection, pre-annotation, LLM or

local-embedding classification suggestions, human correction, provenance and export for more than one domain. It also makes a crucial semantic distinction: uploaded raw text contains no hidden gold labels. Gold comes either from an imported annotated dataset or from reviewer-approved corrections. Department routing is therefore stored separately as model prediction and approved decision.

This proves functional completeness, not annotation-time savings. The largest remaining empirical gap is the human-in-the-loop workflow itself. A valid study should use unseen, representative internal tickets in a randomized, counterbalanced comparison of manual annotation and pre-annotation review. At least two domain experts should independently annotate an overlapping subset, with disagreements adjudicated under a frozen guideline. The application should record active review time excluding idle periods, additions, deletions, boundary changes and relabeling. Primary outcomes should include time per document and final strict quality against adjudicated labels; agreement should be reported with an appropriate chance-corrected statistic and pairwise span scores [34]. A pilot should inform power and sample-size planning. Acceptance rate alone is insufficient because a reviewer can accept wrong but plausible suggestions.

Classification is less validated than extraction. The 1,000-ticket comparison uses synthetic data, while the local embedding route and retrieval-augmented LLM route have no independent target-domain result. A target-ticket study should freeze the queue schema and partition data by customer, conversation thread or time into non-overlapping reference, development and held-out test sets. The reference set may populate the embedding index, the development set may select the prompt and retrieval settings, and the held-out set should be scored once. LLM and embedding routes should be compared on the same tickets using accuracy, macro- F_1 , per-queue confusion, abstention or low-confidence coverage, latency and token use. Gold queue labels must be assigned before reviewers see either prediction.

Engineering recommendations

- 1. Begin with an explicit schema and a small, independent gold set.** Use label definitions, positive and negative examples, and exact boundary rules; do not treat well-formed JSON as evidence of correctness.
- 2. Report strict and reviewer-oriented metrics together.** Strict spans measure training-data readiness, while boundary-tolerant and overlap scores measure whether pre-annotation points a human to the right phrase.
- 3. Treat corrections as data.** Hold out part of the reviewed corpus for evaluation, retrieve approved examples for prompting, and consider fine-tuning only after the annotation policy and inter-annotator agreement are stable.
- 4. Make long runs restartable and cost-visible.** Checkpoint per sentence, hash the prompt and manifest, record token use, and configure actual provider prices before presenting cost. A stored zero with zero rates means unknown, not free.

8.1 Limitations and threats to validity

External validity. Few-NERD is Wikipedia, SciREX is scientific prose, and the ticket corpus is synthetic. None establishes accuracy on an organization's internal tickets or

service reports. All LLM results use one hosted deployment, so model rankings cannot be generalized. SciREX's released mentions combine automatic and human annotation and can contain guideline or annotation noise.

Internal and statistical validity. Most configurations were run once at temperature 0; deterministic decoding reduces but does not eliminate provider-side variation. The 20-paper held-out interval is wide, and the development prompt comparison used only ten detected sentences per window. The 100-paper operational sample is drawn from the training split and is a reliability and scale gate, not a second held-out test. The 1,000-window corpus contains overlapping windows from 438 papers, so window-level independence would be false; this is why intervals cluster by `doc_id`. The full 1,000-window inference run is prepared but incomplete.

Metric validity. Boundary-tolerant and overlap matching are deliberately more forgiving and should never replace strict scoring in claims about gold-data quality. Conversely, strict F_1 understates pre-annotation utility when the reviewer can repair a nearby boundary quickly. Neither measure evaluates time saved, cognitive load, or missed cross-sentence relations.

Product scope. The application is a reproducible single-instance research product. It lacks authentication, role-based permissions, concurrent work allocation, a central database, tenant isolation, and direct integration with a ticketing platform. Reviewer corrections are logged but do not automatically retrain a model. Those controls are required before deployment to hundreds of users or processing confidential tickets.

9 Conclusion and Outlook

We built a reproducible route from raw text to evidence about an LLM system: a configuration-driven evaluation framework, a document-scale SciREX pipeline, and a human-in-the-loop annotation product. The central finding is not that one prompting variant wins universally. It is that structural validity can coexist with poor semantic accuracy, prompt definitions materially change behavior, and quality must be measured on the target distribution.

The fresh SciREX result—0.5738 strict and 0.7030 overlap F_1 —supports pre-annotation followed by review, not autonomous annotation. The 100-paper run shows that sentence-local calls can reliably cover complete windows of more than 200 sentences with checkpointing, while making the scale visible: 5,868 calls and 2.91 million tokens for 100 papers. Reporting such operational evidence beside accuracy is necessary for a defensible deployment decision.

Next steps. The immediate, feasible research steps are to instrument active annotation time and editing actions, pilot a randomized manual-versus-pre-annotation study, and use that pilot to determine the required sample size. The study should use at least two domain experts, overlapping assignments, adjudication and unseen target-domain tickets; it should measure final annotation quality and agreement as well as time, rather than treating acceptance as correctness. In parallel, the LLM, local nearest-neighbor and retrieval-augmented classifiers should be evaluated on the same leakage-safe reference/development/test partition of independently labeled internal tickets.

Next, improve Dataset/Metric definitions and test sentence groups where cross-sentence context matters. The prepared 1,000-window SciREX study should only be launched after actual provider pricing is configured; final train, development and official-test results must be reported separately with paper-clustered intervals.

Operational deployment is a larger engineering and governance phase, not a small extension of the current Streamlit application. It requires organizational authentication, project-scoped roles, concurrent assignment and version control, a transactional database, background jobs, immutable audit logs, backup and recovery, and tenant isolation. Before confidential tickets are processed, the organization must also approve the complete data flow, model endpoint and retention policy; enforce encryption, least privilege and managed secrets; and verify provider data-use terms. These controls require the target organization's infrastructure and security review and cannot be established by repository tests alone. Fine-tuning is not a priority until a sufficiently large, stable and consistently reviewed corpus exists. The Docker distribution and YAML use-case layer make controlled research pilots reproducible without claiming readiness for hundreds of concurrent users.

Artifact Availability

The repository contains task-specific methods, prompts, configurations, preprocessing and evaluation scripts, the 1,000-window SciREX manifest and linked CSV exports, and the Streamlit annotation application. Runtime predictions and score directories are Git-ignored and must be preserved in controlled artifact storage; the report includes only conditions verified against the local retained artifacts. The product is distributed through a pinned Dockerfile and Compose configuration, with a Dev Container as an optional development environment. Synthetic demonstration tickets are included. Credentials, local environment files and reviewed gold exports are excluded from both Git and the Docker build context.

Bibliography

- [1] D. Xu, W. Chen, W. Peng, C. Zhang, T. Xu, X. Zhao *et al.*, "Large language models for generative information extraction: A survey," *Frontiers of Computer Science*, vol. 18, no. 6, p. 186357, 2024, doi: 10.1007/s11704-024-40555-y.
- [2] N. Pangakis, S. Wolken, and N. Fasching, "Automated annotation with generative AI requires validation," *arXiv preprint arXiv:2306.00176*, 2023.
- [3] M. V. Reiss, "Testing the reliability of ChatGPT for text annotation and classification: A cautionary remark," *arXiv preprint arXiv:2304.11085*, 2023.
- [4] S. Jain, M. van Zuylen, H. Hajishirzi, and I. Beltagy, "SciREX: A challenge dataset for document-level information extraction," in *Proc. ACL 2020*, 2020, pp. 7506–7516, doi: 10.18653/v1/2020.acl-main.670.
- [5] X. Wang, W. Zhou, C. Zu, H. Xia, T. Chen, Y. Zhang *et al.*, "InstructUIE: Multi-task instruction tuning for unified information extraction," *arXiv preprint arXiv:2304.08085*, 2023.
- [6] W. Zhou, S. Zhang, Y. Gu, M. Chen, and H. Poon, "UniversalNER: Targeted distillation from large language models for open named entity recognition," in *Proc. ICLR 2024*, 2024.
- [7] U. Zaratiana, N. Tomeh, P. Holat, and T. Charnois, "GLiNER: Generalist model for named entity recognition using bidirectional transformer," in *Proc. NAACL 2024*, 2024, pp. 5364–5376, doi: 10.18653/v1/2024.naacl-long.300.
- [8] R. Han, T. Peng, C. Yang, B. Wang, L. Liu, and X. Wan, "Is information extraction solved by ChatGPT? an analysis of performance, evaluation criteria, robustness and errors," *arXiv preprint arXiv:2305.14450*, 2023.
- [9] T. Xie, Q. Li, J. Zhang, Y. Zhang, Z. Liu, and H. Wang, "Empirical study of zero-shot NER with ChatGPT," in *Proc. EMNLP 2023*, 2023, pp. 7935–7956, doi: 10.18653/v1/2023.emnlp-main.493.
- [10] S. Sivarajkumar, M. Kelley, A. Samolyk-Mazzanti, S. Visweswaran, and Y. Wang, "An empirical evaluation of prompting strategies for large language models in zero-shot clinical natural language processing," *JMIR Medical Informatics*, vol. 12, p. e55318, 2024, doi: 10.2196/55318.
- [11] S. Wang, X. Sun, X. Li, R. Ouyang, F. Wu, T. Zhang, J. Li, G. Wang, and C. Guo, "GPT-NER: Named entity recognition via large language models," in *Findings of the Association for Computational Linguistics: NAACL 2025*. Albuquerque, New Mexico: Association for Computational Linguistics, 2025, pp. 4257–4275, doi: 10.18653/v1/2025.findings-naacl.239.
- [12] W. Yin, J. Hay, and D. Roth, "Benchmarking zero-shot text classification: Datasets, evaluation and entailment approach," in *Proc. EMNLP-IJCNLP 2019*, 2019, pp. 3914–3923, doi: 10.18653/v1/D19-1404.
- [13] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in *Proc. EMNLP-IJCNLP 2019*, 2019, pp. 3982–3992, doi: 10.18653/v1/D19-1410.
- [14] L. Tunstall, N. Reimers, U. E. S. Jo, L. Bates, D. Korat, M. Wasserblat, and O. Pereg, "Efficient few-shot learning without prompts," *arXiv preprint arXiv:2209.11055*, 2022.
- [15] N. Muennighoff, N. Tazi, L. Magne, and N. Reimers, "MTEB: Massive text embedding benchmark," in *Proc. EACL 2023*, 2023, pp. 2014–2037, doi: 10.18653/v1/2023.eacl-main.148.
- [16] R. Bhambhoria, L. Chen, and X. Zhu, "A simple and effective framework for strict zero-shot hierarchical classification," in *Proc. ACL 2023 (Short Papers)*, 2023, pp. 1782–1792, doi: 10.18653/v1/2023.acl-short.152.
- [17] Y. Zhang, R. Yang, X. Xu, R. Li, J. Xiao, J. Shen, and J. Han, "TELEClass: Taxonomy enrichment and LLM-enhanced hierarchical text classification with minimal supervision," in *Proc. WWW 2025*, 2025, doi: 10.1145/3696410.3714940.
- [18] Q. Zhang, Q. Su, W. Zhu, and Y. Pang, "HierPrompt: Zero-shot hierarchical text classification with LLM-enhanced prototypes," in *Findings of EMNLP 2025*, 2025, pp. 3846–3859.
- [19] Z. R. Tam, C.-K. Wu, Y.-L. Tsai, C.-Y. Lin, H.-y. Lee, and Y.-N. Chen, "Let me speak freely? a study on the impact of format restrictions on large language model performance," in *Proc. EMNLP 2024, Industry Track*, 2024, pp. 1218–1236, doi: 10.18653/v1/2024.emnlp-industry.91.
- [20] J. Huang, X. Chen, S. Mishra, H. S. Zheng, A. W. Yu, X. Song, and D. Zhou, "Large language models cannot self-correct reasoning yet," in *Proc. ICLR 2024*, 2024.
- [21] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang *et al.*, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in *Proc. NeurIPS 2023, Datasets and Benchmarks Track*, 2023.
- [22] F. Gilardi, M. Alizadeh, and M. Kubli, "ChatGPT outperforms crowd workers for text-annotation tasks," *Proceedings of the National Academy of Sciences*, vol. 120, no. 30, p. e2305016120, 2023, doi: 10.1073/pnas.2305016120.
- [23] C. Ziems, W. Held, O. Shaikh, J. Chen, Z. Zhang, and D. Yang, "Can large language models transform computational social science?" *Computational Linguistics*, vol. 50, no. 1, pp. 237–291, 2024, doi: 10.1162/coli_a_00502.
- [24] J. Baumann, P. Röttger, A. Urman, A. Wendsjö, F. M. Plaza-del Arco, J. B. Gruber, and D. Hovy, "Large language model hacking: Quantifying the hidden risks of using LLMs for text annotation," *arXiv preprint arXiv:2509.08825*, 2025.
- [25] Z. Tan, D. Li, S. Wang, A. Beigi, B. Jiang, A. Bhattacharjee *et al.*, "Large language models for data annotation and synthesis: A survey," in *Proc. EMNLP 2024*, 2024, pp. 930–957, doi: 10.18653/v1/2024.emnlp-main.54.
- [26] J. Geng, F. Cai, Y. Wang, H. Koeppl, P. Nakov, and I. Gurevych, "A survey of confidence estimation and calibration in large language models," in *Proc. NAACL 2024*, 2024, pp. 6577–6595, doi: 10.18653/v1/2024.naacl-long.366.
- [27] S. Kadavath, T. Conerly, A. Askell, T. Henighan, D. Drain, E. Perez *et al.*, "Language models (mostly) know what they know," *arXiv preprint arXiv:2207.05221*, 2022.
- [28] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," in *Proc. ICLR 2023*, 2023.
- [29] L. Kuhn, Y. Gal, and S. Farquhar, "Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation," in *Proc. ICLR 2023*, 2023.
- [30] M. Xiong, Z. Hu, X. Lu, Y. Li, J. Fu, J. He, and B. Hooi, "Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs," in *Proc. ICLR 2024*, 2024.
- [31] X. Wang, H. Kim, S. Rahman, K. Mitra, and Z. Miao, "Human-LLM collaborative annotation through effective verification of LLM labels," in *Proc. CHI 2024*, 2024, doi: 10.1145/3613904.3641960.
- [32] J.-C. Klie, M. Bugert, B. Boullosa, R. Eckart de Castilho, and I. Gurevych, "The INCEPTION platform: Machine-assisted and knowledge-oriented interactive annotation," in *Proc. COLING 2018, System Demonstrations*, 2018, pp. 5–9.
- [33] H. Kim, K. Mitra, R. Li Chen, S. Rahman, and D. Zhang, "MEGAnno+: A human-LLM collaborative annotation system," in *Proc. EACL 2024, System Demonstrations*, 2024, pp. 168–176.
- [34] R. Artstein and M. Poesio, "Survey article: Inter-coder agreement for computational linguistics," *Computational Linguistics*, vol. 34, no. 4, pp. 555–596, 2008, doi: 10.1162/coli.07-034-R2.
- [35] T. Bueck, "Multilingual customer support tickets," Hugging Face dataset `Tobi-Bueck/customer-support-tickets`, 2024, revision `9469f9edfb311e578514c31cf58b9b42939029c5`; source file `aa_dataset-tickets-multi-lang-5-2-50-version.csv`; accessed 26 July 2026.
- [36] N. Ding, G. Xu, Y. Chen, X. Wang, X. Han, P. Xie, H. Zheng, and Z. Liu, "Few-NERD: A few-shot named entity recognition dataset," in *Proc. ACL-IJCNLP 2021*, 2021, pp. 3198–3213, doi: 10.18653/v1/2021.acl-long.248.

Appendix A Additional Embedding-Based Ticket Routing Experiments

A.1 Zero-Shot Embedding Similarity Classification

To complement the prompted ticket classifiers, we implemented a training-free semantic label-matching baseline. A ticket text x and every allowed class label c are encoded by the same pretrained sentence encoder f . The predicted class is the label whose vector has the highest cosine similarity to the ticket vector.

$$\begin{aligned} e_x &= f(x), & e_c &= f(c) \\ c_{\text{hat}} &= \operatorname{argmax}_c \cos(e_x, e_c) \end{aligned}$$

The procedure is applied independently to the ticket type label set and to the queue label set. It requires no LLM call, no task-specific training, and no retrieval of labeled reference tickets. Because the output label is selected directly from the allowed schema, malformed generative output is structurally impossible. This method therefore isolates how far representation geometry alone can solve the routing task.

A.2 Evaluation on the 1,000-Ticket Slice

The embedding-similarity method was evaluated on the same fixed 1,000-ticket slice used for the short-input ticket baselines. The run achieved type accuracy 0.505 with type macro-F1 0.446, and queue accuracy 0.270 with queue macro-F1 0.242. These values are lower than the hosted zero-shot LLM baseline reported in Table 1, but they quantify a useful cost-accuracy trade-off: fully local deterministic inference remains possible without prompt construction or API usage.

Method	Type Acc.	Type mF1	Queue Acc.	Queue mF1
Zero-shot LLM	.726	.606	.422	.366
Agent two-step	.719	.632	.420	.363
Embedding similarity	.505	.446	.270	.242

A.3 Dynamic Embedding RAG

A second embedding-based route uses dense retrieval to select in-context examples for an LLM. Reference tickets are encoded once. At inference time the new ticket is embedded in the same vector space, cosine similarity is computed against the reference index, and the top- k semantically closest valid tickets are inserted into a dynamically constructed few-shot prompt. The LLM then predicts the structured ticket type, queue, and tags.

The implementation uses query-time exclusion of an identical ticket ID or normalized text as an additional leakage safeguard. This differs from static few-shot prompting because the examples change for every target ticket and are selected by semantic proximity rather than being hard-coded.

A.4 Reporting Status of the RAG Result

The dynamic retrieval method is implemented and was exercised during development, but the retained synthetic-ticket experiment did not provide a clean independent reference/development/test partition for the retrieval index. For that reason, its development scores are not promoted into the verified comparison in Table 1. A defensible final benchmark should build the retrieval index only from a frozen reference split, tune k and prompt choices on development data, and score the held-out test split once.

A.5 Interpretation

The added experiments separate three distinct sources of capability. The embedding-similarity baseline measures pure semantic geometry; the zero-shot and agentic baselines measure direct generative classification; and embedding RAG combines local retrieval with generative reasoning. The lower queue score of pure similarity is consistent with the difficulty of matching long ticket narratives directly to short department names, while its deterministic local execution makes it a useful lightweight baseline.

These results therefore extend, rather than replace, the original short-input comparison. They show that embeddings can either act as the classifier itself or as a retrieval layer that supplies context to an LLM, with different trade-offs in accuracy, cost, latency, and evaluation requirements.

Note: The embedding-similarity metrics above come from the completed local 1,000-ticket run. The dynamic RAG method is described as an implemented exploratory method; no non-independent RAG score is treated as a verified benchmark result.